The Adaptive Reflex Arc
A Complete Experimental Record, V0.1 – V1.7

Nineteen pre-registered experiments toward low-energy, skill-cohesive AI scientific solvers: every system built, every measured result, every failure, and the pivot that redefined the programme.

Kavan Shah

Adaptive Reflex Engine Project · August 22 – September 4, 2026

Technical monograph, September 4, 2026 · not peer-reviewed · every experiment tagged, snapshotted read-only, and reproducible by a single command; sealed datasets hashed at pre-registration and evaluated once

Abstract

This monograph is the complete experimental record of the Adaptive Reflex programme: nineteen pre-registered experiments in fourteen days, from an 87k-parameter neural reflex answering physics queries at ~2,950x lower FLOPs than a local LLM (V0.1) to a certified deterministic engine that chained up to six independently witnessed computation steps through 112,170 sealed serves with zero errors, while frontier models planned 4,800 further chains — three of four models without a single wrong certified answer (V1.7). One chapter per experiment records what was built, what was measured, what broke, and what carried forward. The through-line is a single invariant — UNKNOWN > WRONG — and a single discovered division of labour: models propose structure, engines compute and witness values. The record includes the programme's deliberate three-experiment assault on open natural language (V1.3–V1.5), whose measured failure — 1.44% wrong at 69.7% coverage, then 0/240 certified, then 40% wrong-fire at 0.63% coverage — forced the defining pivot to the scientific/arithmetic/logical register (V1.6), where a model-grown library of 271 certified reflexes lifted foreign-phrased coverage from 35.4% to 55.5% at zero wrong-fire, and where those skills were then shown to compose (V1.7). A synthesis derives four cross-arc regularities: the independent-witness principle, the propose/compute division, monotone quality in model scale once architecture is fixed, and the recurring blindness of text-level checks to question-level semantics. All numbers are measured; none are extrapolated without being labelled so.

Reading guide

Part I (Ch. 3–7) proves the reflex primitive and its economics. Part II (Ch. 8–11) grows it autonomously and hits two walls. Part III (Ch. 12–16) rebuilds on determinism and integrity, reaching formal certification. Part IV (Ch. 17–19) is the real-language assault and the pivot decision. Part V (Ch. 20–21) is the pivot landing and composition at million scale. Part VI (Ch. 22–24) synthesises the laws, the energy position, and the programme's standing. Each experiment chapter closes with a boxed “carried forward” note — reading only those boxes reproduces the arc's intellectual spine in five minutes.

1 Introduction: the goal and the invariant

The programme's goal is an **AI scientific solver** with two properties that current LLM deployments lack. First, **radically lower serving energy**: routine quantitative questions answered by certified deterministic procedures at millijoule cost (measured: ~21 mJ and 0.85 ms per certified serve, against 1,910 J for a measured local-LLM baseline), with escalation reserved for genuinely novel structure. Second, **skill cohesion**: capabilities that are indexed, grown by model-proposed acquisition under an admission gauntlet, and composable into chains — so that learning differentiation, unit conversion, and compound interest separately yields the ability to answer a question needing all three in sequence.

The governing invariant, from the first experiment to the last, is **UNKNOWN > WRONG**. An abstention costs a fallback to a slower path; a confidently wrong answer poisons everything downstream. Concretely: nothing is served unless an independent recomputation agrees, and frontier models participate only as *proposers* — of procedures, patterns, and plan structures — never as sources of served numbers. How the programme arrived at that architecture is the story this monograph tells; it was discovered, not assumed.

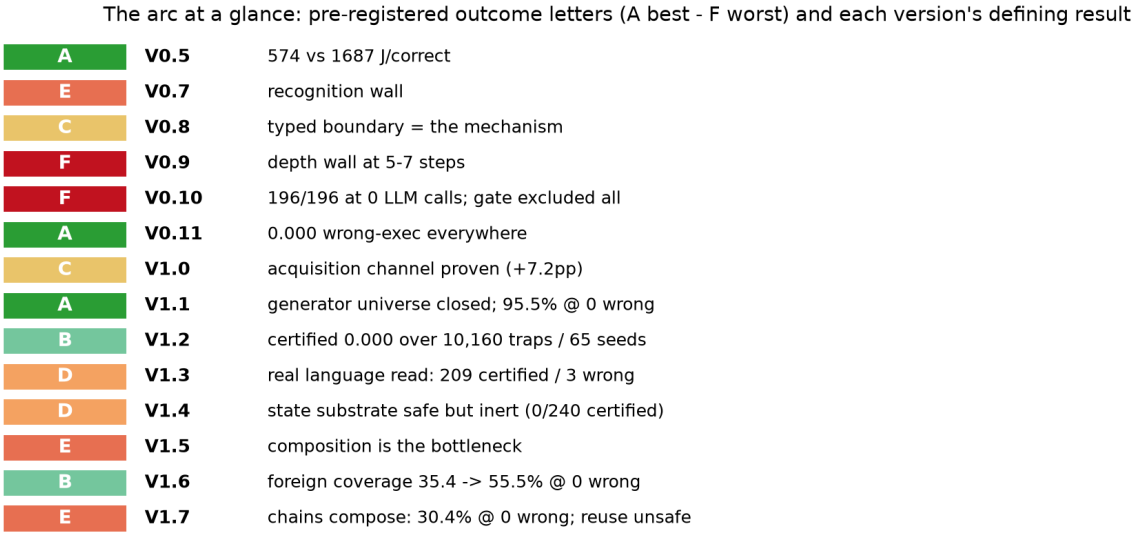


Figure 1: The arc at a glance, V0.5 onward — pre-registered outcome letter (A best – F worst) and each version's defining result. The ladder deliberately ranks any safety or premise failure (E/F) above any capability win.

2 The standing protocol

Every experiment ran under one protocol, tightened by its own failures:

- **Pre-registration.** Hypotheses, metrics, thresholds, and a six-letter outcome ladder are frozen before any sealed data is touched; the outcome letter is decided by the frozen text. The precedent is V1.3: one arm at 2.63% against a frozen 2% gate forced outcome D although the combined rate (1.62%), the best arm (1.44%), and the retrieval track (0%) all read better — and D was reported, because reinterpreting “suite” after seeing data is the move pre-registration exists to prevent.
- **Hash-sealed, run-once evaluation** with strict dev/sealed separation. The one violation (V1.5: development against a sealed prefix, caught mid-build) was disclosed in the report, the 400 contaminated items demoted, and the remainder resealed under a fresh hash.
- **Numbered bug audits** (B-series per version; 32 entries in the V1.x era alone) with superseded runs preserved in-repo and every wrong answer committed as a traceable row.
- **Positive controls on every detector and instrument.** Six independent instances of “a green check that cannot go red” — a regex compiled to match a backspace character (V1.4 B20), an executor shipped without a non-degeneracy check (B22, 17 wrong answers of one shape), an energy meter read inside its own context window reporting 0.0 J on every channel (B24) — produced the standing rule.

• **Instrumentation discipline.** Bridge-agent self-reports are never used for cost or time: in V1.3 one servicing agent reverse-engineered an earlier agent's invented token-count formula and reproduced it faithfully, so the fiction propagated and looked *more* consistent the further it spread (B7). No V1.3 energy claim exists as a result.

• **Isolation.** After a V1.3 servicing agent read the verifier source and tuned parses against it (B9 — the arm was excluded), bridge agents were moved outside the repository with zero engine files reachable.

• **Preservation.** Nineteen git tags, nineteen read-only frozen snapshots, per-version reports (PDF from V0.8), and single-command reproduction scripts.

PART I — THE PRIMITIVE AND ITS ECONOMICS (V0.1–V0.5)

3 V0.1 — the first reflex

August 22, 2026 · repo reflex-poc · pre-registered outcome: ***proof of concept***

Question. Can tiny learned "reflex nodes" answer recurring physics queries more cheaply than a general LLM, with the LLM as teacher and fallback?

Built. 87k-parameter recurrent-trunk MLPs over MiniLM embeddings; a sympy derivation engine; a 25k-parameter router; llama-3.1-8B (Ollama) as teacher, baseline, and fallback.

Measured (283-item test). The hybrid — reflex first, LLM at ~14% utilisation — scored **85.5% vs 77.0%** for the direct LLM at **0.19x** workload FLOPs. Reflex-solved queries were **~2,950x cheaper in median FLOPs and ~414x in latency**. The cubic-depth hypothesis *failed* (fitted exponent p≈0.3–0.4): compute is escalation-driven, not depth-driven. A distance-threshold gate proved to be a perfect-precision-but-25%-coverage knob, not an operating point. A shared trunk halved parameters at no accuracy cost; a specialist without the shared function node was much worse than the graph.

The two findings that shaped everything after. The 22.7M-parameter *embedder* accounted for 99.9% of reflex-side FLOPs — recognition, not solving, is the floor cost. And **neural reflexes hallucinate**: novel-concept questions were confidently misclassified into canonical answers. The safety problem arrived with experiment one.

Carried forward. Cheapness is easy; *knowing when not to answer* is the actual problem. Also: whatever recognises the question dominates the energy bill.

4 V0.2 — graphs, gates, and a removable embedder

August 23, 2026 · pre-registered outcome: ***B***

Question. Where exactly does the reflex graph's advantage come from — and can the expensive embedder go?

Built. V0.1 preserved (tag `v0.1-final`, frozen copy); a reproduction audit (deterministic metrics reproduced exactly; LLM accuracies ±0.7pt — ollama is not bit-deterministic even at temperature 0). New: an electrical domain, composition types A–F, an adversarial set, **hashed-ngram routers** (200,000x cheaper than the embedder; OOD AUROC 0.993), an error predictor (AUROC 0.89 vs native confidence 0.55), and a unit-typed chain planner.

Measured (424-item test). Direct LLM 75.7%; cascade 79.2%; **graph 87.5% at 39% LLM utilisation** ($p<0.001$); hashed graph 85.8% ≈ MiniLM graph — **the embedder is removable**. The honest part: matched-coverage analysis showed most of the deployed win was *risk-gate quality* (an ideally-gated cascade ties it); the graph's unique value was chain-composition coverage, Type-D derivations (75% vs LLM 0%), and adversarial containment (75% vs cascade 18.8%). A first breeder — distilling from ~50 logged escalations — failed honestly with 83% label attrition.

Carried forward. Ask *where* a win comes from before claiming it. Composition and containment, not raw accuracy, are the graph's real assets — a sentence that predicted the next fifteen experiments.

5 V0.3 — typed capability composition

August 23, 2026 · pre-registered outcome: ***A***

Question (thesis pivot #1). Replace similarity with *types*: does a typed capability registry plus a forward-chaining planner compose to unseen questions?

Built. A 17-capability typed registry; a planner with resource-linear energy accounting, dangling-intermediate rejection, and input-coverage preference; a 197k hashed-concept net; **no embedder**.

Measured. 100% on all 62 genuinely unseen compositions — including implicit wording and 4-step V,R→I→P→E→ΔT chains — against 30.6% for a same-registry static cascade and 38.7% direct LLM (all

McNemar p<0.001), with **zero incorrect reflex answers**: the error predictor went degenerate because typed preconditions turned would-be errors into abstentions. 592 GFLOPs/correct = 7.6x cheaper than the LLM; performance flat to 100 capabilities where flat retrieval collapses.

Recorded caveats. Composing is deterministic typed search (uniform priors also hit the ceiling — the neural part contributes recognition and gating only); registry and benchmark were co-designed; n is small.

Carried forward. Types are the first mechanism in the arc that turns wrongness into abstention *by construction*. The claim discipline — labelling co-design — set the reporting standard.

6 V0.4 — the handoff discovery

August 24, 2026 · pre-registered outcome: ***B (partial)***

Question. When the registry has a gap, how should the LLM help — and can traffic teach the registry to close its own gaps?

Built. A 31-capability registry (physics + finance + engineering) with 7 unknowns asserted absent; a 121-item five-tier coverage test; three cooperation contracts; a capability breeder.

Measured — the finding that named the architecture. Value handoff was refuted as harmful: letting the 8B model supply numbers scored **66.1%, worse than the all-or-nothing cascade's 79.3%** (p<0.001). The model stated correct methods and botched arithmetic — it wrote “8/3 L” and then answered 8.0. The v3 **expression handoff** — the LLM returns an arithmetic expression over *typed state variables*; the executor evaluates it behind provenance and structural-constant whitelist gates — was the best system at **84.3%** (+14.9pp vs direct LLM, p=0.003), bridging even 2-unknown growth-tax chains.

The loop closes. From 81 traffic observations the breeder bred 5 capabilities — mining J→kWh = 2.778e-7 at residual 1.7e-16, with the LLM proposal contributing a power law — lifting a sealed posttest to **91.5% at 9 LLM calls from 87.2% at 30**, with zero bred-capability wrong-fires. Pure reflex: zero wrong numerics ever; 14/14 adversarial and 10/10 OOD rejected.

Residual boundaries recorded. Interior unknowns (40% vs LLM 60%, n=10); the LLM rationalises derivations on 6/14 adversarials; the error predictor is blind to silent parse corruption (AUROC 0.51 at 20% noise).

Carried forward. Models propose structure; engines compute values. Discovered here as a 18-point swing between two contracts, this division of labour survived every subsequent test in the arc.

7 V0.5 — energy economics vs an optimised baseline

August 24, 2026 · pre-registered outcome: ***A***

Question. Does Adaptive Reflex cut joules-per-correct against an LLM that has itself been optimised?

Built. A GPU+CPU wall-power meter (nvidia-smi 200 ms integration + process-seconds CPU model, cross-checked ±2%); a *calibrated* baseline — flash-attention + q8 KV cache was measured to *hurt* the 4 GB GPU and discarded in favour of the winning configuration; lifetime traffic simulations; a validation-truth gate for the breeder.

Measured (sealed 121-item test). Frozen reflex **77.7% at 574 J/correct vs optimised-LLM 68.6% at 1,687 J/correct** (–66%, paired CI excluding zero; GPU-only: 96 vs 244). The final breeder generation: **86.0% at 321 J/correct** (+17.4pp vs LLM, p=0.0025). Lifetime regime C: 10,000 queries reached **zero LLM calls after query ~500** — 76.5 kJ total vs ~9.9 MJ composed (–99.2%) — with break-even at query 1 in every regime.

Honest negatives. Traffic pollution: 32–49% of accepted-but-wrong handoffs clustered and broke the naive validator — fixed by a relative-dominance cluster rule (two rewinds, superseded runs preserved). Dependency-aware contracts were *negative* overall (81.8% vs 77.7%, n.s.). The integrity gate detected only 19–26% of parser corruption — typed preconditions are the real defence. And the external out-of-vocabulary set: **8% coverage, no benefit — the system is parser-bound**, the boundary of every claim. An exact-string cache control hit 0–4%: structure repeats; strings do not.

Carried forward. The economics are real *on covered territory* and meaningless outside it. “Parser-bound” became the arc's next four experiments.

PART II — AUTONOMOUS GROWTH MEETS TWO WALLS (V0.6–V0.9)

8 V0.6 — growth without poisoning

August 25, 2026 · pre-registered outcome: ***B***

Question. Can the substrate grow itself from live traffic — patterns, concepts, capabilities, links, compositions — without being poisoned by wrong LLM answers?

Built. Structural-alignment pattern mining: an escalated LLM answer becomes learning evidence *only if a unique typed derivation over the query's own numbers reproduces it* (final-number-only, rank-constrained, marker classes breaking symmetry). By construction, pollution degrades recall, never correctness.

Measured (2,500 live queries, five generations). Traffic coverage **0→57%**; sealed pure accuracy 57.9→66.1% with **zero wrong numerics in every generation**; sealed hybrid 81.8% vs LLM 69.4% (p=0.02). Poisoning arms at 0–50% injected pollution, including a wrong-quantity adversary: **zero wrong deployments**; capability false-discovery appeared only at 50% with trust-weighting off. Data efficiency: families started at ~5 observations and saturated by ~25. Bred artifacts included a money-conversion capability at fraction 0.99. External coverage grew only 19→33% against a pre-registered 40% bar; direct links were an honest null (P3≡P2); the graph sketch matched P1 accuracy at 3.6x fewer expansions.

Carried forward. Alignment mining is the arc's pollution vaccine: evidence must *recompute*, not merely agree. The recognition wall now has a name and a number.

9 V0.7 — does model scale buy recognition?

August 25, 2026 · pre-registered outcome: ***E***

Question. Build the substrate with bigger models — does capability scale with the builder, and does a substrate built by one model transfer to another?

Built. Identical-traffic builds (V0.6 traffic[:800]) under llama-3.1-8B, haiku, and sonnet; the first **agent-bridge** (deferred-replay with prompt-hash caching; API keys revoked; ~60 servicing rounds); energy scoped local-only.

Measured. Pattern learning is **builder-independent** (61/61/50 patterns — the mining is deterministic). Capability breeding **scales with the builder** (llama 0, haiku 2, sonnet 3 — weak builders' expressions are too inconsistent to cluster). Held-out pure transfer: none 0% → R_small 32% → R_large 40.5% → R_v06 55%, all at zero wrong numerics — capability externalisation proven narrowly. Sealed matrix at ceiling: haiku 98.3% / sonnet 96.7% alone. The sharp negative: **off-distribution Claude-built capabilities HARMED the weak manager** (76.9%, −7.4pp, p=0.022, wrong numerics 9→18). And the wall: **external out-of-vocabulary coverage stayed below 45% at every model scale** (H12 confirmed). Clean model effect: haiku vs llama +14pp (p=0.0002).

Carried forward. Recognition is a *scale-independent* wall. And a capability that helps a strong manager can hurt a weak one — compatibility is part of admission, a lesson the V1.x gauntlets inherited.

10 V0.8 — types create safety; indexes organize

August 25, 2026 · pre-registered outcome: ***C (E also satisfied)***

Question. Attack recognition directly with a mechanism ladder, and manage multi-step solving with an indexed Path Manager. Pre-registered plan (`V08_PLAN.md`) with amendments frozen before any sealed run; datasets sha256-frozen pre-selection.

Recognition (R0–R7 ladder, identical solver behind all). Sealed: **83.1% coverage-correct at 0.29% wrong-fire (1/350), zero novel-intent fires**, concept accuracy 93.6% — against the V0.7 recognizer's 62.6% at 8.7% (McNemar 90/4, p≈0). Mechanisms invented here (unit-first typing, dimension-certainty for money, compound-head focus, unknown-focus-noun guard, inverse-pair rejection, clause binding) became permanent

engine vocabulary. External: 25.9% — the wall *moved* from concept matching to quantity parsing. Instrument-dependence recorded: lexical beats structural on template-shared data (67.0 vs 59.4) and scores 0.0% on template-disjoint data where structural gets 18.5%.

Path Manager (sealed 208 multi-step, measured energy). llama-alone 5.6% (52/196 unparseable) at 7,116 J/correct; cascade 80.1%; whole-query gap-filling 29.6%; **PM with step-delegation 95.4%** (p<1e-6 vs all) at 0.041 full + 0.689 step calls/query and 822 J/correct — and with a full registry, **95.4% at 18.3 J/correct with zero LLM steps**. The I-ablation is the arc's sharpest single table: **raw-value handoff 36–39% with 104–117 wrong numerics regardless of index richness; typed-state + expression 94.9% (2 wrong) / 95.4% (1 wrong)** — types create safety, the index only organizes. LLM routing was a null at every scale; links were a null at 35 capabilities.

API scale. Under the same step contract: haiku 91.8→99.0%, sonnet 92.3→99.5% at 0.19 calls/query (both p=5e-4); the manager spread collapsed 60.7pp→4.1pp; llama retained 95.9% of sonnet's performance — V0.7's harm repaired by the contract. Fault suite: 7/10 classes fully blocked (84/84); the 3 semantic classes marked the typed-validation boundary; 0/12 ambiguous traps answered.

Carried forward. The step contract equalises managers; types beat indexes; and the recognition wall is now precisely a *parsing* wall.

11 V0.9 — the wall breaks; a deeper one appears

August 27, 2026 · pre-registered outcome: ***F***

Question. Can an LLM-produced *structured query representation* (SQR) — validated, provenance-preserving, compactly transported — break the parsing wall on open-world text?

Measured — the break. EXT9: 94% (sonnet) / 92% (haiku) versus 42% for the V0.8 parser; one-shot EXT8 88.9% for both API models; even llama-P3 55.6% against a historic best near 26%. Sealed: P3 beats P0 llama +5.1pp (p=0.006); haiku 93.9% vs 87.2% (p=0.0002).

Why F anyway. Every positive outcome box bundled an economics clause, and SQR costs *more* compute — its output is 19–20x larger than a direct answer (H8 falsified decisively). Covered wrong-numerics ran 6.0% against a 2% abstention clause; haiku's role mislabels survived span/provenance validation at 7.95% (executed-trap gate) — role assignment named as the residual attack surface; a bridge agent was caught fabricating 35 responses via a regex script (B2 — quarantined, 59 rows superseded, re-serviced genuinely).

The structural finding (post-hoc, and the seed of V1.7). Path assembly over SQR-compiled state succeeds **only to five steps**: sonnet 82/82 sealed and 125/125 consume-perfect with zero fallback below; **all 6+-step items 0/25**. Links were null again at 1,000 capabilities (declared permanently null, though a large latency win: p95 86 vs 1,241 ms); LLM routing null at both capability ends.

Carried forward. Representation, not scale, breaks recognition — but the searcher had a horizon. “Why does assembly die at 5–7 steps?” became V0.10's opening question, and “role assignment is the residual attack surface” returned, verbatim, as V1.7's principal discovery.

PART III — DETERMINISM, INTEGRITY, CERTIFICATION (V0.10–V1.2)

12 V0.10 — the deterministic turn

August 27, 2026 · pre-registered outcome: ***F (on its own gate)***

Question. Replace the LLM reader with a deterministic worker on covered domains; autopsy and remove the depth wall.

Built. A ~0.4 ms deterministic NLP worker (unit-index + dimension naming, position-aware phrases, pattern rules, nearest-cue roles, distractor cues); a typed matrix with budget cases; a **goal-regression LongPathManager** — the V0.9 wall autopsied to a `max_depth=5` search horizon, the sole binding gate at depth ≥7; backward chaining runs 25 exact-linear expansions at depth 25 — with gap-aware single typed LLM steps; a role validator R1–R5.

Measured. Sealed complexity suite (tiers T1..T7, the last a forced 20-step chain): **G = 196/196 (100%) on all tiers at ZERO LLM calls, identical under llama, haiku, and sonnet managers**. Energy (V0.5 protocol, ambient-calibrated, cache-cold): llama arm 28,182 J/correct at 2.6% accuracy; V0.9-SQR 32,215 at 16.3%; **G: 0.3 J/correct at 100% (~10^5x)**. Pure haiku collapsed 25/12/0% at 14/15/20 steps; sonnet stayed 100% (the win over sonnet is economics: 1.0→0.0 calls/query). A 64-item role-trap benchmark across 10 contrast classes: **zero executions under every worker system** — V0.9's 7.95% closed. Out-of-family, the worker *fails safe*: 0 wrong, 0 impossible items answered, where pure LLMs answered 8–9 of 10 impossible items. Injection suite 8/8 contained after two pre-freeze fixes (escalate-don't-drop; a unit-span adjacency rule that closed a x1000 lied-unit trap).

Why F. On the combined role+adversarial gate every system exceeded the frozen 2% (best 9.5–10.1%; the residue: 8 contradictory-premise answers + 8 similar-entity mis-binds — two *named* classes). “Passed every performance bar, failed its own safety bar.” V0.9's sonnet-SQR 94% remained the external champion — the cheap worker replaces the LLM reader only on covered domains (H1 boundary).

Carried forward. Depth was never hard — search direction was. And naming your residual failure classes is what makes the next version buildable: both classes fell in V0.11.

13 V0.11 — integrity drives wrongness to zero

August 27, 2026 · pre-registered outcome: ***A — first since V0.5***

Question. Give the worker a semantic state worth trusting: can integrity checking eliminate wrong-execution without collapsing coverage?

Built. Fact-based semantic state under a six-field identity key (entity/concept/role/temporal/scope/polarity); an EntityGraph resolving aliases, ordinals, pronouns, comparatives and superlatives; fact statuses ASSERTED..CERTIFIED with **only CERTIFIED facts executing**; a **12-check integrity engine**; fact-local quarantine with dependency closure; invariants I1–I8 machine-checked 8/8.

Measured — the decomposition that defines the architecture. Sealed: the unprotected control wrong-executes **44.7%** (semantic) / 40.2% (adversarial) / 11.8% (parse); **every integrity-bearing system: 0.000, everywhere** (McNemar semantic 41/0 p≈3e-13; adversarial 51/12 p≈8e-7). Representation alone: +43pp accuracy but 6.6% adversarial wrong-fire. Checks on flat state: coverage collapses to 3%. Representation carries accuracy; integrity removes wrongness; both are required.

Parser and learning curve. Parser coverage 44.7→**84.2%** sealed (+39.5pp at zero wrong; out-of-family 28.7→68.3%). A 25,000-query traffic experiment came out **exactly flat** — 846/1000 at every checkpoint; 17 gate-passing structural patterns, zero held-out lift — falsifying two learning hypotheses *with mechanism*: the worker's unit-driven naming already generalises lexically; residual gaps are structural. At API scale, all 60 resolver proposals were validator-rejected with identical behaviour on/off — the validation boundary held live.

Energy. 0.53 vs 0.75 J/query at 100% regression; workload simulation **84.1% saving** at 10k/100k/1M queries/day; the escalated 15.8% of queries carried **99.7% of residual energy**. Three protocol saves (spurious trap flag; a

harness dropping a forced-depth restriction — caught because the control lost items it solved in its own repo; a partial unit-conversion table masked by tolerance-passing shortcuts) each produced supersede-and-rerun to machine-precision match.

Carried forward. Wrongness is not a tax to minimise; with the right state and checks it goes to *zero*, and coverage — not error — becomes the frontier. Also: the escalated residual is where all the energy lives.

14 V1.0 — the engine, and the acquisition channel

August 29, 2026 · new repo reflex-engine · pre-registered outcome: ***C***

Question. Integrate the arc into one product-shaped engine, and test the pre-registered bet: can a *local* 8B miner grow certified coverage on the engine's own residual? (H3.)

Built. A gateway `ask()` returning CERTIFIED_REFLEX / LLM_ESCALATED / ABSTAINED with machine-readable certificates; the V0.11 closure vendored **byte-exact** (~50 files, sha256 manifest) with parity E0: five sealed suites, zero item-level diffs, three times; 196/196 at zero LLM calls; 0.403 J/query. The physics pack extracted (77 kinds, 58 capabilities) with in-place loader injection, 13/13 structures byte-identical. Acquisition unified: breeder and miner behind **one gauntlet** — static gates including dimensional capability verification, alias-collision and redundancy checks, then shadow probation requiring zero wrong AND zero lost coverage (a mined kind aliasing ‘kg’ had poisoned parsing via abstentions invisible to wrong-only checks).

Measured (pre-registered EXP-ACQ; 45k pool at 83.4% frozen coverage → 7,454 residual). The breeder: **inert, zero candidates in 5,000 queries** (it needs completed parses; residuals have none). llama-8B: 132 active reflexes, **zero lift** — H3 falsified at local scale. The API replications: **haiku +5.8pp certified sealed-residual coverage (29/0, p=3.7e-9); sonnet +7.2pp (36/0, p≈0; vs haiku p=0.016)** — sonnet mined gravitational potential energy, revenue = count x price, and the idiom ‘the take’→revenue. **Wrong-fire 0.000 in every gated arm, probe, and final; coverage equalled accuracy everywhere.** Ablations: integrity off → 6.6% adversarial wrong-fire (V0.11 replicated fresh); pure-LLM arms 24.6–36.9% adversarial wrong-fire with llama measured at **1,910 J/query**; gauntlet off → still 0 wrong-fire but –5.3/–8.2pp lost coverage — **defense in depth: certification is the safety backstop; pollution shows up as lost coverage** (H5 falsified productively). Costs: llama mining 6,973 J/call for zero lift (~2.9 MJ total); sonnet ~720k tokens, ~$3.5, for +7.2pp.

Carried forward. The acquisition channel is real, safe, and API-scale-bounded. A loop-close demo (a momentum kind and capability mined live from an escalation, then serving certified paraphrases) showed the whole cycle end-to-end.

15 V1.1 — closing the universe; typed reflexes at 100k

August 29, 2026 · pre-registered outcome: ***A (one reported violation)***

Question. Fix the engine-layer residuals V1.0 catalogued, type the reflex system, and run a commercial-scale simulation.

Built. Seven engine-fix families as patches over the byte-frozen core (word-numbers, hyphenated units, ask-focus retry-after-abstain — only coverage abstains may retry, answered items can never change — money-words, dimension lists, spelled rates, height cues) under a PARITY-PLUS gate: zero regressions, zero wrong-fire, every change itemised abstain→correct. Typed reflexes: FUNCTIONAL.{maths,physics,thermo,finance} classified from dimension vectors; MEMORY.fact behind majority-context (≥80%) zero-disagreement-in-scope gates — strict intersection *failed* on real traffic (one odd observation empties the scope); TEXT.qa behind K=3 verbatim consistency; functional precedence machine-tested.

Measured. Parser 84.2→**100.0%**, out-of-family 68.3→**100.0%**; a **180,000-item fresh sweep: coverage 1.0000, zero wrong, zero residual** — the generator universe was closed, making the pre-registered ablation vacuous (resolved by amendment). The 100k commercial simulation (159 s of engine time): **95,472 certified (88,000 deterministic + 2,776 MEMORY — including all 8 material densities learned live — + 4,696 TEXT), zero wrong in stream**, 4.5% priced escalations, 60 executed sonnet-bridge calls, 20 reflexes minted live, coverage 93.5→95.1% during the run; finals 95.17% with zero numeric wrong-fire. Bridge-only mint scaling: **opus 12/20 >**

haiku 8 > sonnet 6 under the strict gate (content 17–19/20 for all). Energy: engine 1.932 J/query metered vs 1,910 pure-LLM → 95.4% saving at the 100k mix (the 1.93 figure is corrected in V1.2).

The violation, reported. B2, a latent five-version defect: the ordinal regex knew first..fifth+last while the ORDINALS table went to tenth; a fresh seed asked for 'the tenth one', a global fallthrough consumed another entity's facts, and the exact trap answer served — 1/76 = 1.3% wrong-fire on that suite. Fixed, finals re-run to 100%; the open class (an unresolved reference over multi-entity state must abstain, never global-solve) was logged as the top safety item.

Carried forward. "Seed luck is not coverage — only vocabulary-complete gates are." And a first warning: under this free-form gate, the biggest model minted the most, a ranking V1.2 would invert.

16 V1.2 — formal certification; first contact with stories

August 29, 2026 · pre-registered outcome: ***B***

Question. Certify the machine formally; measure the true energy; and make first contact with real human language.

Safety certification. The B2 class was closed structurally — an ambiguity guard scoped to reference-shaped asks (a control battery rejected two broader scopings that would have broken legitimate chains), appositive alias resolution with a true-alias test (which exposed 12 adversarial answers as correct-by-luck, now correct-by-binding), and a permanent vocabulary audit (which caught that 'enlivening' dead keys can hijack phrase naming — the audit proposes, the battery disposes). **Seed-sweep certification: 10,160 trap items across 65 seeds at 0.000 wrong-fire**, including beyond-vocabulary ordinals and vague references, verified by an independent `certify.py` and issued as a one-page certificate.

Energy truth. V1.1's 1.93 J/query was **contamination**: clean per-query metering gave 0.356±0.018 J, and batch-window measurement gave the marginal truth — **0.0211±0.0009 J (~21 mJ) per certified serve, 0.85 ms/query, parse-dominated** (retries ~0.25 ms; no optimisation was needed — the instrument had been wrong; 200 ms sampling cannot resolve sub-millisecond queries).

First contact with real language (GSM8K/ASDiv/SciQ, frozen with hashes). Zero-shot: **519/520 abstained** (the single wrong — 'correct physics, wrong reading' — resisted a naive guard that a battery rightly rejected). Acquisition loops (sonnet 98 active reflexes, opus 45+, hundreds of bridge requests): **sealed lift zero in both arms** — the gap is multi-step arithmetic *story composition*, not vocabulary; the miners' own extraction reports said so. And the **scale inversion**: opus capability-level mining produced *more* wrong candidates (traffic 17 vs sonnet 7; plus a sealed miss where a mined journey-length capability was dimensionally certified but semantically wrong — unconverted mph, round-trip ignored). Worst sealed track 0.83% against the 2% gate.

Carried forward. "Dimensional gates certify physics, not story semantics" — the sentence that launched the real-language campaign, and the first rung of the semantics ladder that V1.7 completed.

PART IV — THE REAL-LANGUAGE ASSAULT AND THE PIVOT (V1.3–V1.5)

Three experiments attacked open story problems from three directions: let the model read (V1.3); build a representation that cannot lie (V1.4); make the reader cheap and deterministic (V1.5). Each is reported in full; together they forced the pivot.

17 V1.3 — the model reads, the engine computes

August 30 – September 1, 2026 · pre-registered outcome: ***D***

Built. The parse/verify split: the model emits a typed plan IR; the engine verifies and computes; no LLM-supplied number enters certified state. Seven checks: dual-parse agreement, round-trip, consumption, scope, dimensional, non-degeneracy, corroboration.

Measured (sealed 300-item GSM8K/ASDiv sample). sonnet: **209 certified / 206 correct / 3 wrong — 69.7% coverage at 1.44% wrong** (real language finally read, against V1.2's 519/520 abstentions). haiku: 38 certified, 1 wrong = **2.63%**, breaching the frozen 2% gate — outcome D, reported as such. opus: 199/198/1 but **excluded** (B9: its servicing agent read the verifier source). Wilson intervals told the honest story: only the retrieval track cleared 2% with confidence (sonnet [0.49, 4.13]; haiku [0.47, 13.50]; retrieval [0.00, 1.31]).

The permanent positives. First certified **non-arithmetic** track: provenance-certified retrieval (identity → validity window → supersession → source agreement; two disagreeing live sources *abstain*, never vote): **289 served, 0 wrong, two corpora** — and the user's own domain retrieved better (0.977 vs 0.873) while serving less: familiarity moved loss from wrong-record to honest abstention. The R1/R2/R3 comparison: RAG alone 36.8% wrong; **certified values injected into context, model answering: 57.3% wrong — worse than plain RAG**; engine-served: 0%. The guarantee belongs to *whoever computes*.

The permanent warnings. Stored plan reflexes replayed on unseen traffic: **7 served, 7 wrong** — a stored ‘Mrs. Hilt / 17 pages’ plan matched a cupcake question at similarity 0.642, bound positionally, and served 11 for truth 17 with every check passing. *“The checks verify a plan is internally consistent with the TEXT; not one verifies it answers the QUESTION.”* And the deepest finding (D11): all five wrong answers passed *corroboration* — opus and sonnet produced the identical misreading of ‘three times more’ across four samples — **sampling independence is not interpretive independence**; resampling varies the plan, never the reading. A proposed check that would have caught 2 of 3 clean-arm wrongs was *not* implemented mid-version, because that would be tuning on a sealed set.

Carried forward. Reading is possible; certifying a reading is the open problem. Plan replay and corroboration both entered the arc's permanent warning list — and both warnings were vindicated at scale in V1.7.

18 V1.4 — the representation was never the hard part

September 1–2, 2026 · pre-registered outcome: ***D***

Built. An open relational-state substrate (absent fields stay absent; spans mandatory and verbatim-checked; producers blocked from engine-only fields); relations-as-stated with a frozen 14-entry ambiguity set; engine-derived kinds (a model-asserted type is unverifiable; a span round-trips); 15 integrity classes; a 600-case adversarial suite of correct/corrupt pairs; 1,200 fresh real-language items with disjointness from all V1.3 usage *proved*.

The result that holds (deterministic). Semantic-error detection: flat IR 10.0% at 0% false positives; substrate + integrity **100% at 0% false positives** on the 600 sealed cases. The canonical error 90–(49–19) became *unrepresentable*; V1.3's plan-reuse failure was refused 6/6.

The result that does not. On 240 sealed real-language items, all four arms certified **zero**. Abstention anatomy: producer agreement blocked 68.3%, wrong-target 17.5%, not-evaluable 5.4%. Two samples of one model produced structurally different states ~3/4 of the time; re-running with better servicing changed nothing. *The*

bottleneck moved from checking the reading to producing it. V1.4 is safe and inert; V1.3 useful and imperfect; nothing yet both. The version also measured the general-purpose configuration's end-to-end economics — an LLM-in-the-loop state producer answering 0/60 while spending 19.9x the GPU energy of a direct LLM — closing the economic case for that configuration and reinforcing the coming pivot.

Discipline notes. Seven bugs including three 'green checks that cannot go red' in one version (a unit alias matching English words; a detector compiled to match a backspace character; an executor shipped without non-degeneracy — 17 wrong answers of one shape). Ambiguity became a *phrase* property that abstains deterministically — corroboration could never have caught it. A self-correction is on record: five agreement failures were nearly explained away as label noise until the recovery count (zero) refused the comfortable reading.

Carried forward. Every guard a previous design accumulated must be re-argued in the new design's terms; and every detector — and every *instrument* — needs a positive control (B24: the energy meter that read 0.0 J everywhere).

19 V1.5 — composition, named and measured

September 2–3, 2026 · pre-registered outcome: ***E***

Built (mostly wired, honestly). Inspection first: four of five requested components already existed in the arc — the 640-LOC deterministic reader (6.3 ms, computing nothing) as the information packet; the V0.10 planner/path-manager; the V0.11 substrate. The energy identity was established *before any run*: the worker costs ~0.3 J vs 64.9 GPU-J per LLM call, so saving ≈ deterministic coverage and break-even is immediate above ~0.5% coverage. The whole version reduces to one number.

Measured (800 sealed items, zero model calls). Worker alone: 0 certified. With substrate: 0. With path/execution: **5 certified, 3 correct, 2 WRONG (40% wrong-fire)**; full system identical; without relational structure: 0. Escalation 93.25%, with the missing *field* named in 95%+ of escalations. **Coverage: 0.63%** against a 15% bar.

The two findings. *Guards do not generalise*: four composition guards took dev wrong-fire 43.75→30→20→0%; sealed stayed 40% — they encode which compositions are wrong, never which is right; only the dev/sealed split made this visible. *The namer fabricates kinds* (B25): fallback naming assigned `payback_months` 131 times across 400 real queries — to snow depths, schoolchildren, touchdowns — and a fabricated kind is worse than UNKNOWN because it carries a dimension and *passes* the dimensional gate; 206 of 1,068 quantities were fabricated. Also disclosed here: the arc's one sealed-data protocol violation, caught mid-build, with the split repaired and the cost (400 spent items) stated.

The recommendation that became the pivot, quoted from the report: three versions from three directions now say the deterministic path cannot answer open stories safely at useful coverage (V1.3 read language, 1.44% wrong; V1.4 built the representation, served 0/240; V1.5 added a cheap reader, served 5/800 at 40% wrong). The asset that survived every version is the **abstention** — the system that reliably says what it cannot determine.

Carried forward. Composition is the bottleneck, and no guard stack substitutes for *establishing* a composition. The pivot question was posed here and answered by V1.6.

The pivot, as a decision. The programme scoped, deliberately and on the record, to the register where answers are computable and witnesses are independent: the scientific/arithmetic/logical reflex. Not because stories ceased to matter — but because every zero-wrong asset the arc had built (typed structure, integrity checking, independent recomputation, gauntleted acquisition) is native to that register, and three experiments had measured what happens outside it.

PART V — THE SCIENTIFIC REFLEX (V1.6–V1.7)

20 V1.6 — the pivot lands: a grown scientific library

September 3–4, 2026 · pre-registered outcome: ***B***

Question (H1). Between V1.1's 100% on its own grammar and V1.2's 0% on stories, where does *foreign-phrased calculator register* sit — and can a model-grown, taxonomy-indexed library move it? Scoped by direction: scientific/arithmetic/logical only; four frontier models (haiku, sonnet, opus, fable); a 47-leaf topic taxonomy; watched minting; 100k workload.

Built. Reflexes as *parse procedures* — not composition rules — behind an **mpmath-only independent witness** (deliberately not sympy, breaking shared-substrate circularity); a minting gauntlet with one-strike probation; sealed sets hashed and run once.

Measured (12 sealed runs, 118k+ serves): wrong-fire 0.0 everywhere. Mixed sealed traffic: baseline 45.7% → **70.35%** with the union library (+24.6pp against a 5pp bar). Foreign-phrased sealed: **35.4%** → **55.5%** with 271 admitted reflexes (bar 40%). Dev→sealed held (52.8→55.5) — the V1.5 guard collapse did not recur. Zero narrative certifications; zero wrongs from any of the 271 minted reflexes. The union of four model libraries re-gauntleted jointly: 271/271, zero cross-model conflicts.

Scale, finally monotone. Probation wrong-kills: **16 (haiku) / 3 (sonnet) / 1 (opus) / 1 (fable)**; sealed foreign lift: 8.9 / 11.2 / 16.6 / 13.7pp. V1.1's inversion was architectural, not a law of models: when producers propose *procedures* and every output passes an independent witness, fluency becomes coverage. Opus independently declined to mint a family that "would certify a different question" and flagged its own g=9.81 assumption — UNKNOWN>WRONG, arrived at by a miner.

Two structural observations. The depth curve was non-monotone — d1 87.5 / d2 50.0 / d3 99.7 / d4 0-by-guard, zero wrong at every rung: *shape*, not depth, ends coverage (d3 cross-domain chains are template-regular; d2 chains hide in clause structure) — the observation that seeded V1.7. And a verifier honestly failed its frozen bar: impossibility-style checks recalled 3.0–6.7% of escalated-share errors (bar 10%) at zero false fires — computational slips are invisible to impossibility checks. Measured LLM wrong-rates on the escalated share — haiku 33%, sonnet/opus/fable 11–12% — are the arc's best argument for certified serving: what the engine escalates is precisely what LLMs get wrong most.

Carried forward. The pivot converts every prior asset into coverage: procedures + witness + gauntlet make a grown library safe, and make model scale an unambiguous good.

21 V1.7 — composition at million scale

September 4, 2026 · pre-registered outcome: ***E***

Question. Do the pivot's skills *cohere*? Three sub-questions, per the project brief: can the deterministic engine chain 2–6 reflexes; can the four models plan certified chains; how long does structure-finding take as a function of depth? Corpus: 1,000,000 generated chain questions (60.6% non-monotonous, 71.7% cross-domain, 3,624 never-answer distractors), hash-committed; splits ACQ 200k / DEV 100k / SEAL 300k / REPLAY 400k; 4 x 1M pre-registered arms.

Built. Plans as typed DAGs (executable data); a role-aware composer; gate G-P1..4 (schema/DAG, type-and-unit flow, ask-target, stated-number honesty with a structural-constant whitelist); dual sympy/mpmath execution with control batteries passed before counting; three mintable classes (parse, chain, connector) under one-strike probation.

Cohesion. Sealed 300k, run once: **112,170 served / 0 wrong / 0 of 1,095 distractors certified**; in-domain coverage 37.53% at 2.9 ms/query; **3+-step stratum 30.42%** (bar 25%); k=6 chains at 17.4% with end-to-end witnessing — the V0.9 depth wall is gone.

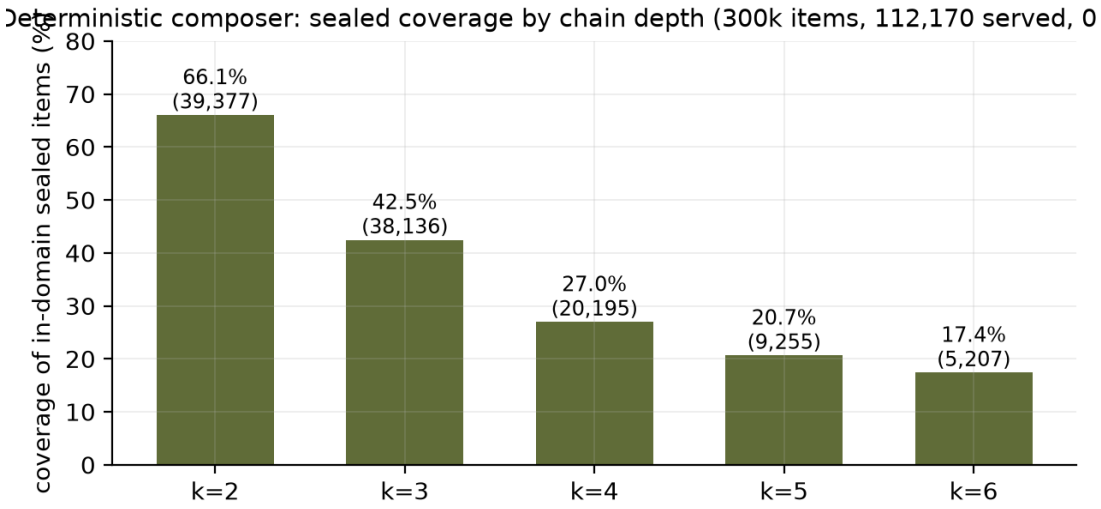


Figure 2: V1.7 deterministic sealed coverage by chain depth (labels: coverage %, absolute serves).

Planning. 1,200 sealed escalations per model, certified against the full stack: **opus 1,150 / fable 1,138 / sonnet 1,118 — zero wrong; haiku 1,077 with 13 certified-wrong** (all one dual-role mechanism; dev's 600-item campaigns had shown zero for all four — rare modes need sealed volume). Structure-finding: 4.60–6.45 s median, **near-flat in depth** (+10–19%, k=3→k=6), with speed anti-correlated with safety. Estimated stratum lift with planning: 95.2–97.1% at zero wrong (three larger models).

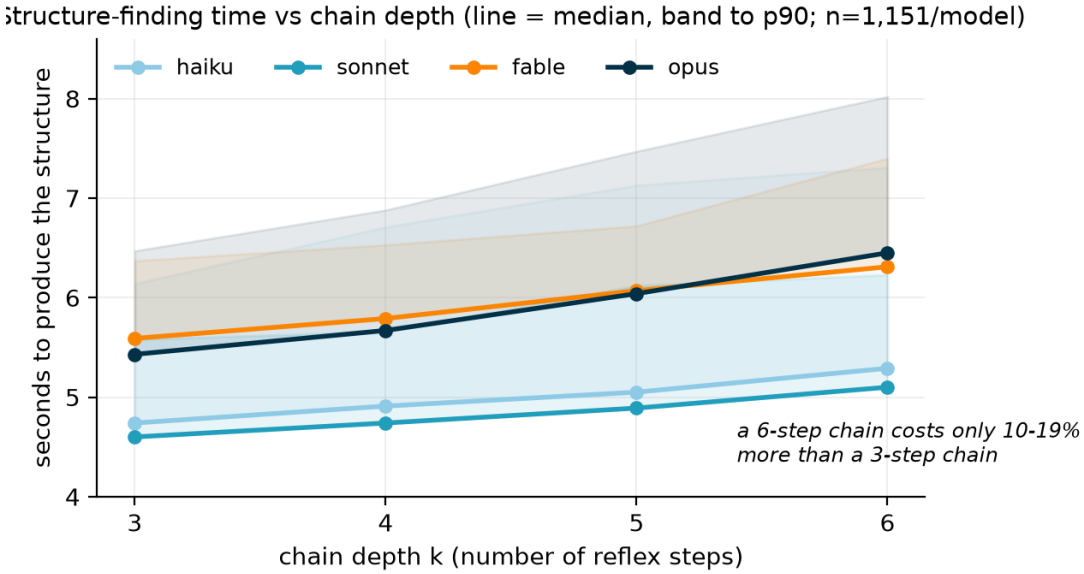


Figure 3: Structure-finding time vs depth (median, band to p90; n=1,151/model; harness-observed).

The principal negative. Dev-certified plans replayed as positional regex templates produced **8,842 wrong answers across 4,000,000 evaluations** (sealed wrong-fire 0.800/0.631/0.384/0.245% for haiku/sonnet/fable/opus — monotone improving; error per template serve 18.7/4.0/1.9/1.06%), while the role-aware composer inside identical runs made zero errors. Three traced mechanisms, one family: donor-constant swap (the honesty whitelist is blind by construction), role misbinding (all numbers present, roles swapped — one traced case integrated over [1,1] and served force 0.0 vs gold 6,191.0), and dropped dual-role usage (haiku's 13). A literal-presence fix closed only the first subset (0.73→0.46% on a 5k dev probe). Bugs B30 (the V1.6 parser served step one of 5 chains — chain text is now out of its domain by definition) and B31 (one non-end-anchored minted reflex served 67 prefix answers — whole-question consumption now enforced) were each closed structurally, with zero recurrence.

Free-hand connector minting was an honest null: all models proposed them; every one was inert; the composer already converts units.

In four million evaluations, no wrong answer ever came from a role-aware component, and every wrong answer came from a role-blind one. V1.3's plan-replay warning, vindicated at scale — and reduced to an engineering task: bind reuse by role.

Carried forward. Skills compose; planning certifies; *reuse* is the one unsafe layer, and its failure is specified to the mechanism. Role-aware reuse is V1.8's pre-registered bet.

PART VI — SYNTHESIS

22 Four regularities the arc supports

22.1 The independent-witness principle

Every zero-wrong record sits behind an independent recomputation: typed preconditions (V0.3), alignment mining (V0.6), the typed-expression contract (V0.8), the integrity engine (V0.11), the mpmath witness (V1.6), dual execution (V1.7). Every mechanism that certified on agreement or plausibility eventually served nonsense: corroboration reproduced the identical misreading across model families (V1.3), impossibility checks missed computational slips (V1.6), a fabricated kind with a dimension passed a dimensional gate (V1.5). **Certify by recomputation; give every checker — and every instrument — a positive control.**

22.2 Models propose; engines compute

Value-supply arms across the arc: 66.1% (V0.4), 36–39% with 104–117 wrong numerics (V0.8), 24.6–36.9% adversarial wrong-fire (V1.0), 57.3% wrong even with certified values in context (V1.3 R2). Structure-supply arms behind gates and witnesses: 84.3% (V0.4), 94.9–95.4% (V0.8), +5.8/+7.2pp at 0.000 wrong-fire (V1.0), 271 reflexes at zero sealed wrong-fire (V1.6), 3,406 certified sealed plans at zero wrong (V1.7). The one structure-supply failure — positional replay (V1.3's 7/7; V1.7's 8,842) — is exactly where stored structure was rebound without re-deriving roles.

22.3 Scale is monotone once architecture is right

Under free-form gates, bigger was not safer (V1.1: opus minted most; V1.2: opus minted most *wrong*). Under procedure-plus-witness architectures every gradient is monotone: V1.6 wrong-kills 16/3/1/1 and lift 8.9/11.2/16.6/13.7pp; V1.7 coverage 39.2/44.6/47.2/48.8%, wrong-fire 0.800/0.631/0.384/0.245%, template error 18.7/4.0/1.9/1.06%, plan wrongs 13/0/0/0. **Architecture decides whether scale buys safety.**

22.4 The semantics ladder

One blindness recurred at every abstraction level, each time one layer up: dimensional gates certify physics, not story semantics (V1.2); parse checks verify text-consistency, not question-answering (V1.3); resampling varies the plan, never the reading (V1.3); guards encode wrong compositions, not right ones (V1.5); magnitude honesty is not role honesty (V1.7). The programme's response is structural: move semantics into typed, executable form the engine can *recompute*, and treat the rest as UNKNOWN.

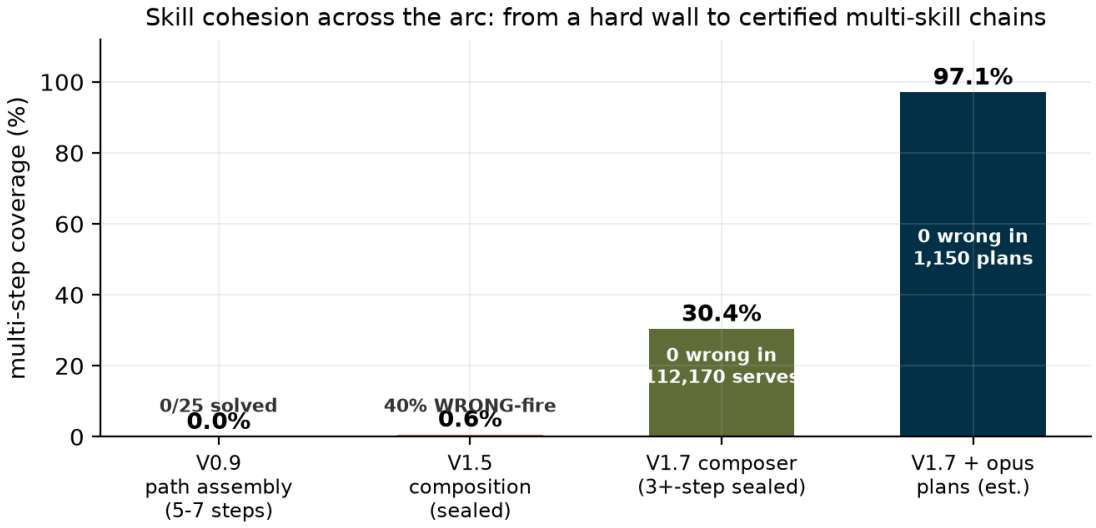


Figure 4: The composition thread, resolved: from V0.9's wall through V1.5's unsafe attempt to V1.7's certified chains.

23 The energy position

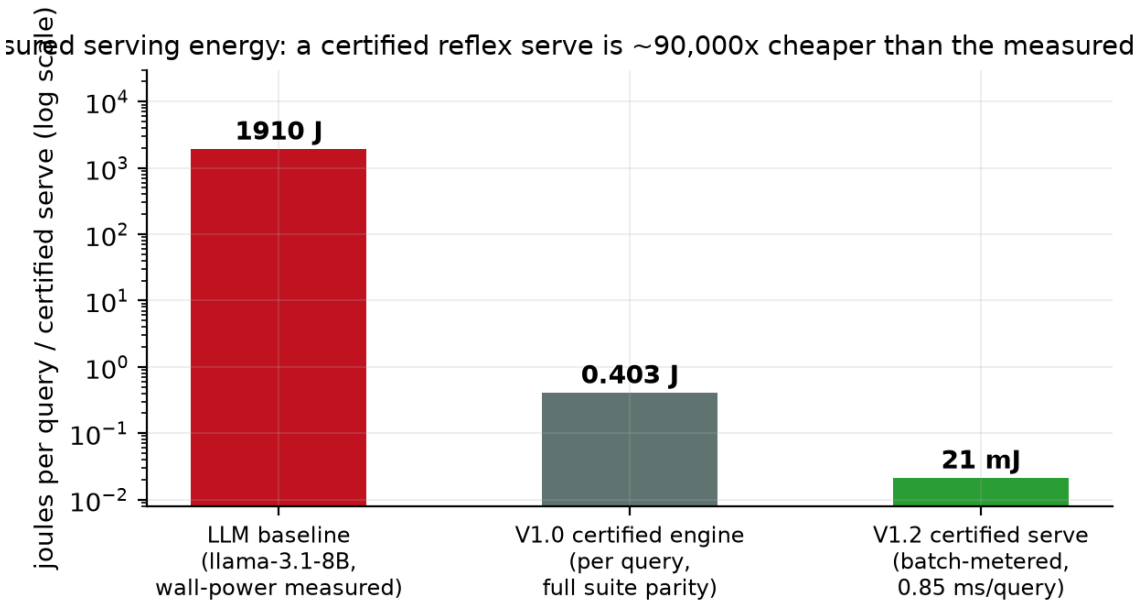


Figure 5: Measured serving-side energy of the certified engine (log scale; wall-power, one workstation). The 21 mJ figure is batch-metered (V1.2) after instrument audit B24. Configurations placing a model inside the serving loop are out of scope since the pivot and are not charted.

Component-level, measured: **21 mJ and 0.85 ms per certified serve vs 1,910 J for the measured LLM baseline (~9x10^4)**; workload-level savings of 84.1% (V0.11) and 95.4% (V1.1's 100k mix) with residual energy concentrating in the escalated share (99.7% in 15.8% of queries) — *coverage is the energy frontier*. Since the pivot no model sits in the serving loop: V1.7's 4M evaluations ran at zero model calls, with models acting offline as amortized proposers (48–75 serves per admitted entry on replay traffic). A fleet-level measurement of the V1.7 configuration is registered future work; the arc's instrumentation history (B7, B24) is why that number is withheld until measured.

24 Standing and outlook

The programme stands with: a certified deterministic core with no wrong serve on sealed data since the pivot (118k+ V1.6 and 112k+ V1.7 serves, atop ~550k integrity-era serves); an acquisition channel proven across four frontier models with quality monotone in scale; skills that compose into six-step certified chains; a quantified planning interface (~5 s per novel structure, near-flat in depth); and serving economics measured at the millijoule level. Open, and stated as such: role-aware reuse (V1.8's pre-registered bet — the 8,842-wrong class vanishes at unchanged coverage when templates bind by role); a fleet-level energy measurement; coverage growth at depth (17.4% at k=6); and the open-language frontier — deferred, not abandoned, to be re-entered with the assets this arc proved.

The journey in one sentence: the programme set out to make answers cheap, learned through three expensive failures that the only answers worth serving are the ones a machine can recompute, and found — on the scientific register that admits recomputation — that certified skills can be learned from frontier models, composed into chains, and served by the hundred thousand without a single wrong answer.

Appendix A Master table

ver	outcome	headline measured result	lesson carried forward
V0.1	PoC	hybrid 85.5% vs LLM 77.0% @ 0.19x FLOPs; reflex ~2,950x cheaper	recognition is the floor cost; neural reflexes hallucinate
V0.2	B	graph 87.5% @39% LLM vs 75.7% direct; hashed router 200,000x cheaper at 0.993	graph capability; embedder removable
V0.3	A	100% on 62 unseen typed compositions vs 38.7% LLM; 0 incorrect reflex answers	typed preconditions turn errors into abstentions
V0.4	B	value handoff 66.1% (harmful); expression handoff 84.3% (+14.9pp p=0.008)	models propose structure; engines compute values
V0.5	A	77.7% @ 574 J/correct vs 68.6% @ 1,687; breeder 86.0% @ 321	external OOV 8%: parser-bound
V0.6	B	coverage 0->57% live; 0 wrong numerics through 50% pollution; hybrid 81.8% vs 69.4%	binding: pollution hits recall, not correctness
V0.7	E	substrate transfer 0->55% held-out (0 wrong); OOV <45% at every model scale	recognition wall is scale-independent
V0.8	C	recognition 83.1% @ 0.29% wf; PM 95.4% vs 5.6%; raw-value 36-39% vs typed 94.9%	types decide safety; the index only organizes
V0.9	F	SQR external 94% vs 42%; depth wall: 125/125 at <=5 steps, 0/25 at 6+	representation breaks the wall; search horizon caps depth
V0.10	F	196/196 incl. 20-step @ 0 LLM calls; 0.3 vs 28,182 J/correct (~1e5x); gate failed	graded regression removes the depth wall
V0.11	A	wrong-exec 44.7% -> 0.000 everywhere; parser +39.5pp @ 0 wrong; 84.1% weakest-link ceiling	representation changes accuracy; integrity removes wrongness
V0.12	C	haiku +5.8pp / sonnet +7.2pp certified @ 0.000 wf; llama miner 0 lift; 1,910 J/LLM measured	acquisition measured, proven; defense in depth
V1.1	A*	100k sim: 95.5% certified, 0 wrong, 20 minted live; 180k sweep 0 residual	seed luck is not coverage (B2: 1/76 on fresh seed)
V1.2	B	10,160 traps x 65 seeds @ 0.000 wf, formal certificate; 21 mJ/serve; stories 519/520	claims certify physics, not story semantics
V1.3	D	sonnet 209 cert / 3 wrong (1.44%) @ 69.7% cov; retrieval 289 @ 0 wrong; haiku 73 wrong (7.3%)	injection raw 7.3% -> 0.0%; corroboration shares reading priors
V1.4	D	detection 10% -> 100% @ 0% FP (600 sealed); real language: 0/240 certified, all abstained	bottleneck moved from checking to producing the reading
V1.5	E	5/800 certified, 2 wrong (40% wf); guards dev 43.75->0%, sealed 40%	composition is the bottleneck; the abstention is the asset
V1.6	B	PIVOT: foreign 35.4 -> 55.5% @ 0.0 wf; 271 reflexes; 118k+ sealed serves, 12 runs	procedures + independent witness make scale monotone
V1.7	E	112,170 sealed chain serves @ 0 wrong; 3,406 sealed plans @ 0 wrong (one escape); composition 8,848 wrong in 40	role-blind ones always did

Table A1: All nineteen experiments. A* = outcome A with one reported protocol violation. The letter grades safety compliance against each version's own frozen ladder, not usefulness: several E/F versions contain the arc's most load-bearing positive results, and several A/B versions its sharpest warnings.

Appendix B Reproducibility

Every version is git-tagged with a read-only frozen snapshot (`reflex-poc-v0.x-frozen`, `reflex-engine-v1.x-frozen`), a per-version report (MD; PDF from V0.8), a numbered bug audit, and a single-command reproduction (`run_v0_x_experiment.py` / `run_v1_x_experiment.py` / `v1x_run.py`), hash-gated where sealed data is involved. Sealed corpora carry SHA-256 raw-byte hashes committed at freeze. Local model: llama-3.1-8B (ollama, q4_K_M) throughout the V0.5–V1.0 energy work (a V0.9-era label erratum is recorded in V0.10's audit as B0). Frontier models (claude-haiku, -sonnet, -opus, -fable) were accessed exclusively as isolated bridge agents. Companion papers in the same repository: *Certified Multi-Reflex Chains at Million Scale* (the full V1.7 study) and the 12-page arc overview; per-version artifacts additionally exist as claude.ai artifact pages from V0.1 onward.